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Anthropic AI for Science
Rare Disease Research Grants Program

RE: Institutional letter of support – LEGEND-WWOX, Track 1: Rare Disease, Basic Science Research

Dear AI for Science Review Committee,

On behalf of the WWOX Foundation, I am pleased to confirm our sponsorship and strong institutional support for the LEGEND-WWOX application. This letter sets out why the project matters, what we have observed through an author-independent test, and how the Foundation will contribute to its delivery.

Massimo Marino, the Foundation's Rare Disease AI Research Lead and the developer of LEGEND, will serve as project lead, primary contact and primary Claude user. He has the Foundation's full backing to direct the grant work.

1. Rationale and origin

WWOX-related disorders are ultra-rare. WOREE, the severe developmental and epileptic encephalopathy caused by biallelic loss of WWOX function, affects a small number of children worldwide. The Foundation exists because families facing that diagnosis still find very little waiting for them.

WWOX itself, however, is not an obscure gene. It has been studied intensively for two decades across oncology, Alzheimer's disease and other dementias, metabolic disease, and paediatric encephalopathy. Mechanistic evidence that could help explain what a given variant does to a child's brain often exists only in fragments, scattered across literatures that do not cite one another or share a vocabulary. It has not been integrated systematically at the level of variant, tissue and developmental context.

LEGEND began with a straightforward purpose: to bring that evidence together in a rigorously structured corpus.

2. Evidence of value and scientific discipline

Once the corpus was unified, using it to generate inferences and hypotheses became the natural next step, but under a discipline that does not allow data, inference and hypothesis to blur into one another. That separation is central to the system's scientific value.

I conducted an author-independent, read-only test of the system using a separate host reasoning model, without Massimo operating or guiding the workflow. The repository's integrity and consistency checks passed.

I then ran the structural analysis of a recurrent WWOX missense variant against the bundled AlphaFold-derived structure. The system placed the residue in a high-confidence, tightly packed helical region, making structural disruption from a proline substitution biologically plausible.

Crucially, LEGEND did not convert that plausible mechanism into a therapeutic conclusion. When asked whether the variant was therefore a candidate for proteostasis rescue, it distinguished direct observation from reasonable inference and hypothesis. It identified the experiments needed to separate impaired production, insolubility, degradation and other mechanisms, and concluded that the evidence was sufficient to justify investigation but not to claim pharmacological rescuability.

The same session exposed a limitation: title-level triage can underrate a paper's true scientific importance. This reinforces the need for the proposed combination of automated triage, full-text review, structured evidence records and expert adjudication. LEGEND is not intended to provide autonomous curation.

3. Grant-period plan and Foundation contribution

The Foundation's ambition extends beyond a six-month grant. Our multi-year objective is to work through the available WWOX literature, generate systematic inference across it, extend computational and mechanistic analysis to WWOX variants reported in the published WOREE and SCAR12 literature, and support deeper characterisation wherever a strong rationale emerges.

This is our direction of travel, not a commitment attached to the present application.

Within the grant period, the scope is deliberately bounded: triage of the complete enumerated queue; complete, receipt-backed reading of at least 100 accessible full texts, with the remainder as a quality-gated stretch target; cross-disease comparison; variant classification; and formal evaluation.

Specifically, we intend to:

- seek engagement from the Foundation's Scientific Advisory Board as an independent, adversarial human review layer for selected claims and workflows;
- submit internally reviewed findings to DisMech as review-ready contributions to the public record, subject to Monarch's independent acceptance;
- share reviewed findings with relevant clinicians and researchers where appropriate, to inform further analysis and potential future publications;
- classify WWOX variants reported across the published WOREE and SCAR12 literature by molecular consequence, extending structural analysis to variants for which such analysis is interpretable; and
- explore participation by other rare-disease communities, beginning with communities with which we already have strong ties, so the method can scale beyond a single gene.

The Foundation will contribute patient-community context, appropriate governance input and, where feasible, relevant clinical and scientific introductions. Karla Sanjines, the Foundation's Patient Outreach Coordinator, will support community engagement and liaison with affected families.

For clarity, LEGEND is an openly licensed codebase authored by Massimo Marino. The Foundation sponsors and supports the project but does not own the codebase. All scientific outputs remain subject to provenance checks, expert review and the limitations stated in the application. They are not clinical guidance.

4. Why WWOX is a demanding test case

We want Anthropic to know the WWOX gene and to understand why it is an unusually demanding test case for frontier models.

WWOX is a tumour suppressor, a neurodevelopmental gene, and a node touching metabolism, myelination and neurodegeneration. Its evidence base spans fields that rarely speak to one another. It therefore exposes context transfer across the wrong variant or tissue, exposes unwarranted causal certainty, and rewards calibrated abstention.

If a model can reason reliably here, the method has the potential to travel.

This is a two-way proposition. The Foundation is not only seeking compute. We are offering a hard, well-instrumented scientific problem and an organisation prepared to report honestly where the reasoning succeeds, where it fails, and what human review remains necessary.

On this basis, the WWOX Foundation gives the LEGEND-WWOX application its full and unequivocal support.

Yours sincerely,

[Handwritten signature omitted from the public repository copy]

Johann Mentz

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